

## Functions of Random Variables

```
int getRandomNumber()  
{  
    return 4; // chosen by fair dice roll.  
              // guaranteed to be random.  
}
```

<https://xkcd.com/221/>

## FUNCTIONS OF RANDOM VARIABLES

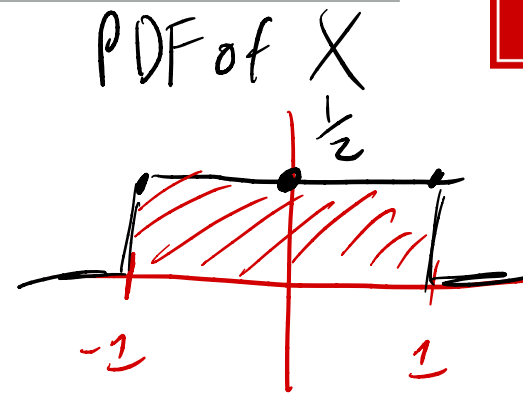
- ▶ A random variable  $X$  is a function that maps outcomes to values:  
 $X: \Omega \rightarrow \mathbb{R}$
- ▶ If we consider a function of  $X$ , such as its square, we get another random variable  $Y = X^2$

## FUNCTIONS OF RANDOM VARIABLES

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 $X: \Omega \rightarrow \mathbb{R}$
- ▶ If we consider a function of  $X$ , such as its square, we get another random variable  $Y = X^2$
- ▶  $Y = X^2$  is shorthand for  $Y(\omega) = (X(\omega))^2$  for all outcomes  $\omega \in \Omega$
- ▶ We can find the range, PDF/PMF, and CDF of  $Y$  from the range, PDF/PMF, and CDF of  $X$

## FUNCTIONS OF RANDOM VARIABLES

- ▶ **Example:**  $X$  is uniform over  $[-1, 1]$
- ▶ Let  $Y = X^2$

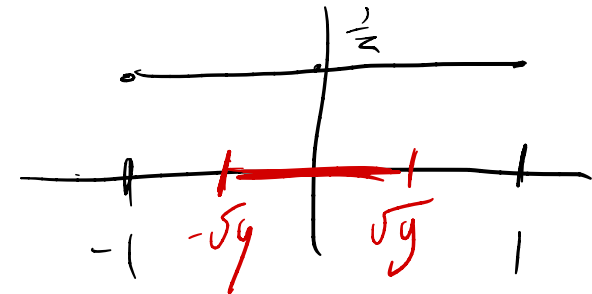


What is the range of  $Y$  (i.e., the set of its possible values)?

$$\text{range}(Y) = [0, 1]$$

## FUNCTIONS OF RANDOM VARIABLES

- ▶ **Example:**  $X$  is uniform over  $[-1,1]$
- ▶ Let  $Y = X^2$



What is the CDF of  $Y$ ?

$$F_Y(y) = \mathbb{P}[Y \leq y] = \mathbb{P}[X^2 \leq y] = \mathbb{P}[|X| \leq \sqrt{y}]$$

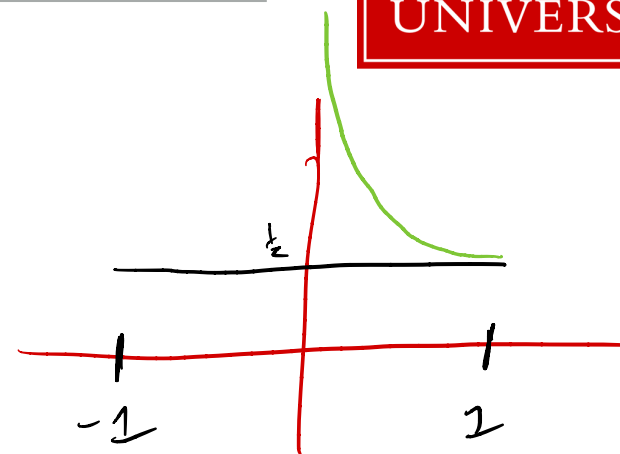
$$F_Y(y) = \begin{cases} 0 & \text{if } y < 0 \\ \sqrt{y} & \text{if } 0 \leq y \leq 1 \\ 1 & \text{if } y > 1 \end{cases}$$

$$= \mathbb{P}[-\sqrt{y} \leq X \leq \sqrt{y}]$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dy = \sqrt{y}$$

## FUNCTIONS OF RANDOM VARIABLES

- ▶ **Example:**  $X$  is uniform over  $[-1,1]$
- ▶ Let  $Y = X^2$



What is the PDF of  $Y$ ?

$$f_Y(y) = F_Y'(y) = \frac{d}{dy} y^{\frac{1}{2}} = \frac{1}{2} y^{-\frac{1}{2}} = \frac{1}{2\sqrt{y}}$$

$$\underline{f_Y(y)} = \begin{cases} 0 & \text{if } y < 0 \text{ or } y > 1 \\ \frac{1}{2\sqrt{y}} & \text{o.w.} \end{cases}$$

## Conditional Probability

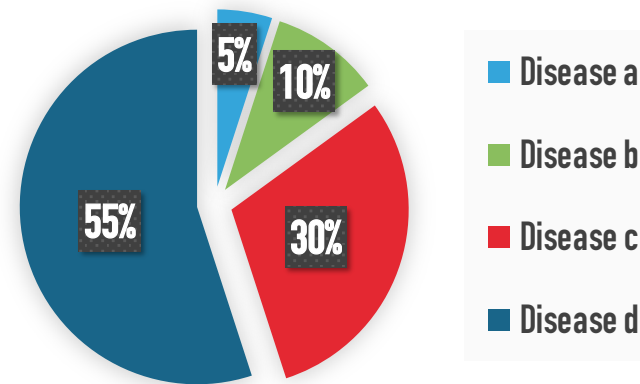


THE ANNUAL DEATH RATE AMONG PEOPLE  
WHO KNOW THAT STATISTIC IS ONE IN SIX.

<https://xkcd.com/795/>

## CONDITIONAL PROBABILITY

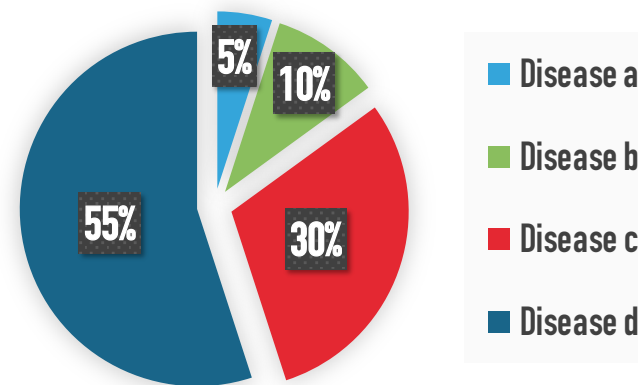
- ▶ A patient has an unknown disease which is one of a, b, c, or d.
- ▶ Based on the symptoms, the doctor estimates that the patient has:
  - ▶ 5% chance of having disease a
  - ▶ 10% chance of having disease b
  - ▶ 30% chance of having disease c
  - ▶ 55% chance of having disease d





## CONDITIONAL PROBABILITY

- ▶ A patient has an unknown disease which is one of a, b, c, or d.
- ▶ Based on the symptoms, the doctor estimates that the patient has:
  - ▶ 5% chance of having disease a
  - ▶ 10% chance of having disease b
  - ▶ 30% chance of having disease c
  - ▶ 55% chance of having disease d
- ▶ The doctor administers a test that reveals that the patient does **not** have disease b or d
- ▶ Based on this information, what is the chance of having disease a?



(A)  $\frac{1}{20}$       (B) 0      (C)  $\frac{1}{10}$       (D)  $\frac{1}{7}$

## CONDITIONAL PROBABILITY

Before the test

| disease | Pr   |
|---------|------|
| a       | 0.05 |
| b       | 0.1  |
| c       | 0.3  |
| d       | 0.55 |

After the test

| disease | Pr  |
|---------|-----|
| a       | ??? |
| b       | ??? |
| c       | ??? |
| d       | ??? |

## CONDITIONAL PROBABILITY

| Before the test |      |   | After the test |     |
|-----------------|------|---|----------------|-----|
| disease         | Pr   |   | disease        | Pr  |
| a               | 0.05 |   | a              | ??? |
| b               | 0.1  | → | b              | 0   |
| c               | 0.3  |   | c              | ??? |
| d               | 0.55 | → | d              | 0   |

The test rules out b and d

## CONDITIONAL PROBABILITY

| Before the test |      |   | After the test |                |
|-----------------|------|---|----------------|----------------|
| disease         | Pr   |   | disease        | Pr             |
| a               | 0.05 | → | a              | $0.05 \cdot x$ |
| b               | 0.1  | → | b              | 0              |
| c               | 0.3  | → | c              | $0.3 \cdot x$  |
| d               | 0.55 | → | d              | 0              |

In the absence of any other information, the new probability of a and c should be **proportional** to their original probability

## CONDITIONAL PROBABILITY

| Before the test |      |   | After the test |                     |
|-----------------|------|---|----------------|---------------------|
| disease         | Pr   |   | disease        | Pr                  |
| a               | 0.05 | → | a              | $\frac{0.05}{0.35}$ |
| b               | 0.1  | → | b              | 0                   |
| c               | 0.3  | → | c              | $\frac{0.3}{0.35}$  |
| d               | 0.55 | → | d              | 0                   |

The probabilities should add up to 1:  $\underline{0.05x} + \underline{0.3x} = \underline{1} \Rightarrow x = \underline{\frac{1}{0.35}}$

## CONDITIONAL PROBABILITY

| Before the test |      |   | After the test |                                   |
|-----------------|------|---|----------------|-----------------------------------|
| disease         | Pr   |   | disease        | Pr                                |
| a               | 0.05 | → | a              | $\frac{0.05}{0.35} = \frac{1}{7}$ |
| b               | 0.1  | → | b              | 0                                 |
| c               | 0.3  | → | c              | $\frac{0.3}{0.35} = \frac{6}{7}$  |
| d               | 0.55 | → | d              | 0                                 |

Note:  $\mathbb{P}(\text{"the disease is neither b nor d"}) = \mathbb{P}(\{a, c\}) = 0.35$

## CONDITIONAL PROBABILITY

- ▶ The information provided by the test is that the following event has happened:

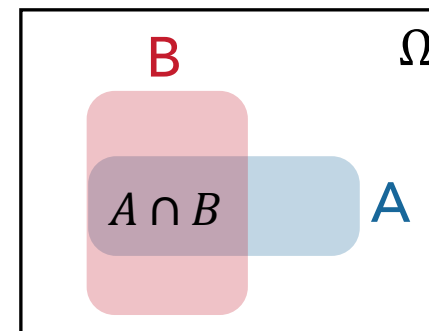
$B = \text{"the disease is neither b nor d"} = \{a, c\}$

- ▶ We defined a new probability function that assigns a probability  $\mathbb{P}(A \mid B)$  to every event  $A \subseteq \Omega = \{a, b, c, d\}$

Conditional  
probability of  
A given B

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Read: probability of A **given** B



## CONDITIONAL PROBABILITY

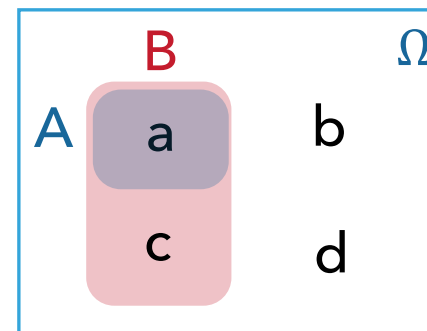
- ▶ The information provided by the test is that the following event has happened:

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$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$\mathbb{P}(\{a\} \mid B) = \frac{\mathbb{P}(\{a\})}{\mathbb{P}(B)} = \frac{0.05}{0.35}$$





## CONDITIONAL PROBABILITY

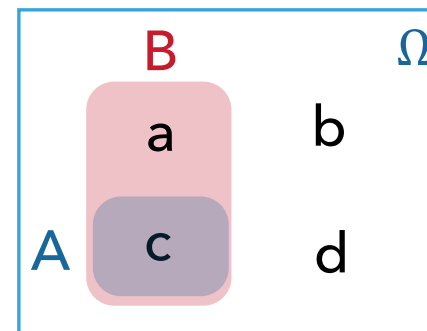
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- ▶ We defined a new probability function that assigns a probability  $\mathbb{P}(A \mid B)$  to every event  $A \subseteq \Omega = \{a, b, c, d\}$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$\mathbb{P}(\{c\} \mid B) = \frac{\mathbb{P}(\{c\})}{\mathbb{P}(B)} = \frac{0.3}{0.35}$$



## CONDITIONAL PROBABILITY

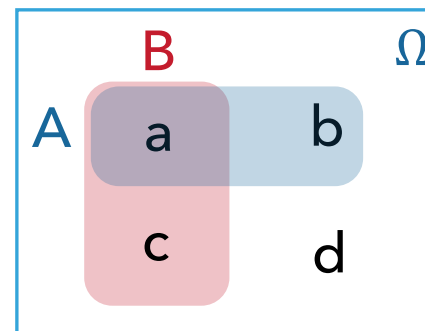
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$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$\mathbb{P}(\{a, b\} \mid B) = \frac{\mathbb{P}(\{a\})}{\mathbb{P}(B)} = \frac{0.05}{0.35}$$



## CONDITIONAL PROBABILITY

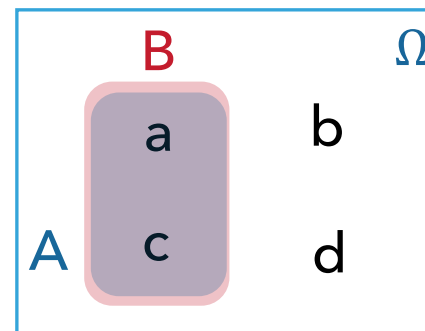
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- ▶ We defined a new probability function that assigns a probability  $\mathbb{P}(A \mid B)$  to every event  $A \subseteq \Omega = \{a, b, c, d\}$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$\mathbb{P}(\{a, c\} \mid B) = \frac{\mathbb{P}(\{a, c\})}{\mathbb{P}(B)} = 1$$



## CONDITIONAL PROBABILITY

- ▶ **Definition:** Let  $B$  be an event such that  $Pr(B) \neq 0$ . For every event  $A$ , we define

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

- ▶  $\mathbb{P}(A \mid B)$  is undefined if  $\mathbb{P}(B) = 0$

## ASK THE AUDIENCE

- ▶ Experiment: roll two fair 6-sided dice
- ▶  $A$  = "first die is 6"
- ▶  $B$  = "sum of the two rolls is 9"

What is  $\mathbb{P}(A \mid B)$ ?

(A)  $\frac{1}{36}$

(B)  $\frac{1}{6}$

(C)  $\frac{1}{4}$

(D) None of the above

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}[\{(6, 3)\}]}{\mathbb{P}[\{(3, 6), (4, 5), (5, 4), (6, 3)\}]}$$

$$= \frac{\frac{1}{36}}{\frac{4}{36}} = \frac{1}{4}$$

## ASK THE AUDIENCE

- ▶ Experiment: toss a fair coin three times
- ▶  $A$  = "more H than T"
- ▶  $B$  = "first toss is H"

What is  $\mathbb{P}(A | B)$ ?

(A)  $\frac{3}{8}$

(B)  $\frac{1}{2}$

(C)  $\frac{3}{4}$

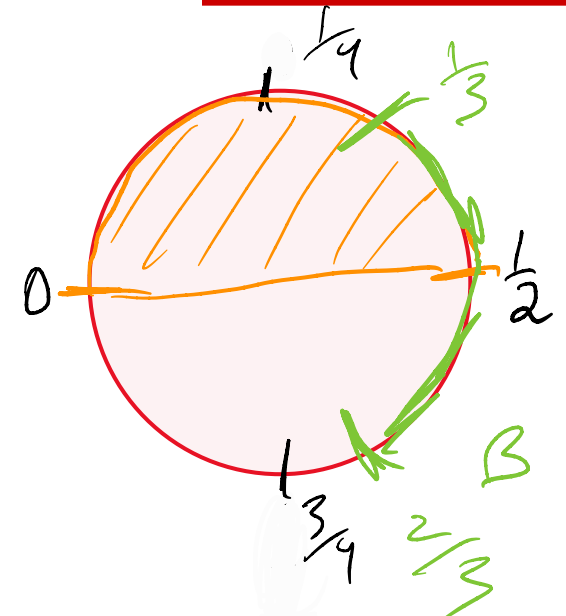
(D) None of the above

know first toss is H  
 so,  $\mathbb{P}[A|B] = \mathbb{P}[\geq 1 \text{ H in 2 tosses}]$   
 $= 1 - \mathbb{P}[0 \text{ H}]$   
 $= 1 - \frac{1}{4} = \boxed{\frac{3}{4}}$

$$\frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]} = \frac{\mathbb{P}[\{HHH, HHT, HTH\}]}{\frac{1}{2}} = \frac{\frac{3}{8}}{\frac{1}{2}} = \boxed{\frac{3}{4}}$$

## ASK THE AUDIENCE

- ▶ Experiment: spin the dial of the spinner
- ▶  $A$  = "dial stopped in the upper half of the circle"
- ▶  $B$  = "outcome is in  $[1/3, 2/3]$ "



What is  $\mathbb{P}(A \mid B)$ ?

(A)  $\frac{1}{3}$

(B)  $\frac{1}{2}$

(C)  $\frac{3}{4}$

(D) None of the above

$$\frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]} = \frac{\mathbb{P}\left[\left[\frac{1}{3}, \frac{1}{2}\right]\right]}{\mathbb{P}\left[\left[\frac{1}{3}, \frac{2}{3}\right]\right]} = \frac{\frac{1}{6}}{\frac{1}{3}} = \frac{1}{2}$$

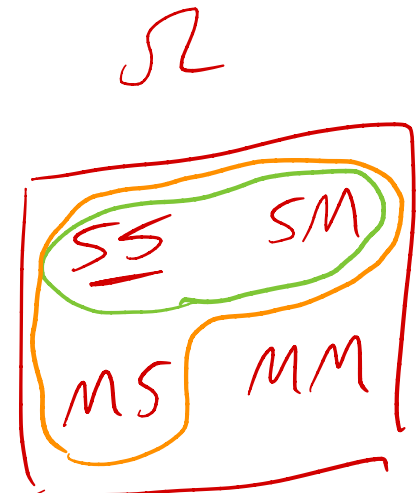
## ASK THE AUDIENCE

- ▶ The Terriers kick two penalties in a soccer game. Each penalty has a 50% chance of scoring a goal.
- ▶ A = "both penalties score"
- ▶ B = "the first penalty scores"
- ▶ C = "at least one penalty scores"

True or False:  $\mathbb{P}(A \mid B) = \mathbb{P}(A \mid C)$

(A) True

(B) False





## ASK THE AUDIENCE

- ▶ The Terriers kick two penalties in a soccer game. Each penalty has a 50% chance of scoring a goal and the outcome of each kick is independent of the other.

- ▶ A = "both penalties score"

True or False:  $\mathbb{P}(A \mid B) = \mathbb{P}(A \mid C)$

- ▶ B = "the first penalty scores"

- ▶ C = "at least one penalty scores"

## CONDITIONAL PROBABILITIES ARE PROBABILITY FUNCTIONS

- ▶ Conditional probabilities  $\mathbb{P}(\cdot | B)$  for any event  $B$  with non-zero probability **satisfy the axioms of probability**
- ▶ **Non-negativity:** for every event  $A$ , we have  $\mathbb{P}(A | B) \geq 0$

- ▶ **Additivity:** if  $A$  and  $C$  are **disjoint** events, we have

$$\mathbb{P}(A \cup C | B) = \mathbb{P}(A | B) + \mathbb{P}(C | B)$$

- ▶ **Normalization:**  $\mathbb{P}(\Omega | B) = 1$
- ▶ Note: it holds only for conditional probabilities  $\mathbb{P}(\cdot | B)$  with the same conditioning event

## CONDITIONAL PROBABILITY: AXIOM PROOFS

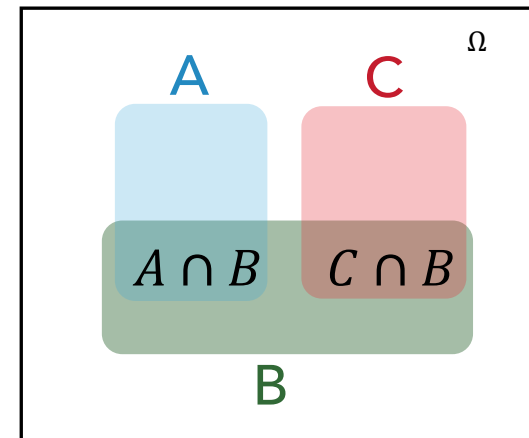
- ▶ Conditional probabilities satisfy the **non-negativity** axiom
- ▶ **Non-negativity**: for every event  $A$ , we have  $\mathbb{P}(A | B) \geq 0$

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \geq 0 \quad \left( \text{by non-negativity of } \mathbb{P}, \text{ and } \mathbb{P}(B) > 0 \right)$$

## CONDITIONAL PROBABILITY: AXIOM PROOFS

- Conditional probabilities satisfy the **additivity** axiom:
- If  $A, C$  are **disjoint**,  $\mathbb{P}(A \cup C \mid B) = \mathbb{P}(A \mid B) + \mathbb{P}(C \mid B)$

$$\begin{aligned} \mathbb{P}(A \cup C \mid B) &= \frac{\mathbb{P}[(A \cup C) \cap B]}{\mathbb{P}[B]} \\ &= \frac{\mathbb{P}[(A \cap B) \cup (C \cap B)]}{\mathbb{P}[B]} \end{aligned}$$



$$\begin{aligned} &\text{(by additivity of } \mathbb{P} \text{)} \\ &= \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]} + \frac{\mathbb{P}[C \cap B]}{\mathbb{P}[B]} = \mathbb{P}[A \mid B] + \mathbb{P}[C \mid B] \end{aligned}$$

## CONDITIONAL PROBABILITY: AXIOM PROOFS

- ▶ Conditional probabilities satisfy the **normalization** axiom
- ▶ **Normalization:**  $\mathbb{P}(\Omega \mid B) = 1$

$$\mathbb{P}[\Omega \mid B] = \frac{\mathbb{P}[\Omega \cap B]}{\mathbb{P}[B]} = \frac{\mathbb{P}[B]}{\mathbb{P}[B]} = 1$$

## CONDITIONAL PROBABILITY RULES

- ▶ Since the conditional probability function **is a probability function**, all the probability rules we have learned remain valid
- ▶ Example: **inclusion-exclusion** rule

**Unconditional:**  $\mathbb{P}(A \cup C) = \mathbb{P}(A) + \mathbb{P}(C) - \mathbb{P}(A \cap C)$

**Conditional:**  $\mathbb{P}(A \cup C \mid B) = \mathbb{P}(A \mid B) + \mathbb{P}(C \mid B) - \mathbb{P}(A \cap C \mid B)$

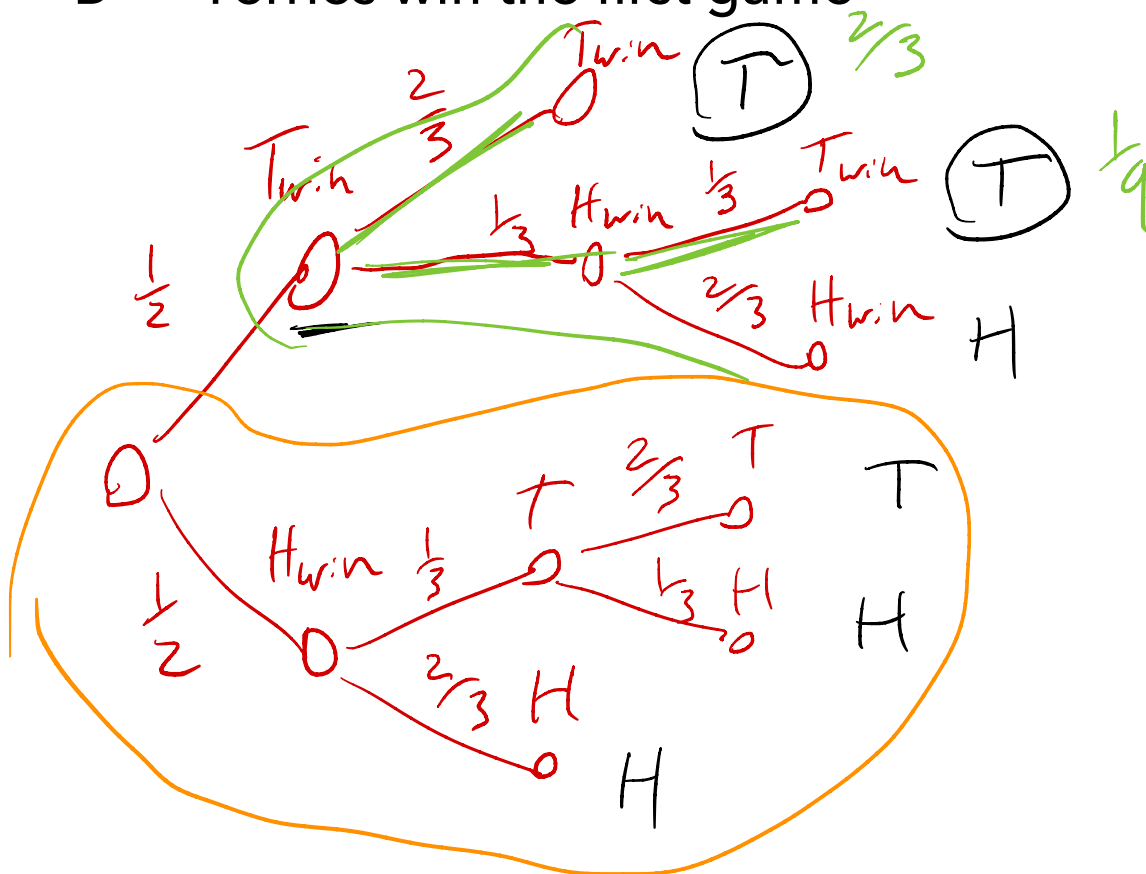
## TREE DIAGRAMS AND THE CHAIN RULE

- ▶ Best-of-three tournament between BU and Harvard's competitive programming teams
- ▶ First game: Terriers win with probability  $1/2$
- ▶ Subsequent games:
  - ▶ If the Terriers won the previous game, they win the current game with probability  $2/3$
  - ▶ If the Terriers lost the previous game, they win the current game with probability  $1/3$
- ▶ What is the probability that the Terriers win the tournament **given** that they win the first game?

$$\begin{aligned} & \text{IP}(\text{win game 3} \mid \text{win game 2}) \\ & \text{IP}(\text{win game 2} \mid \text{win game 1}) \\ & = 2/3 \end{aligned}$$

## TREE DIAGRAMS

- ▶ A = "Terriers win the tournament"
- ▶ B = "Terriers win the first game"

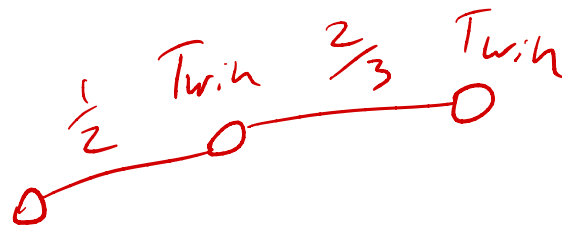


$$P[A|B] = \frac{P[A \cap B]}{P[B]} = \frac{\frac{1}{3} + \frac{1}{18}}{\frac{1}{2}}$$

$$= \frac{\frac{7}{18}}{\frac{1}{2}} = \boxed{\frac{7}{9}}$$

$$\frac{2}{3} + \frac{1}{9} = \boxed{\frac{7}{9}}$$





## CHAIN RULE

- ▶ In the tree diagram, we used the following chain rule:

$$\mathbb{P}(\text{"win game 1 and 2"}) = \mathbb{P}(\text{"win game 2"} | \text{"win game 1"}) \cdot \mathbb{P}(\text{"win game 1"})$$

- ▶ The chain rule follows from the definition of conditional probability:

$$E_1 = \text{"win game 1"}$$

$$E_2 = \text{"win game 2"}$$

$$\mathbb{P}(E_2 | E_1) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_1)} \Rightarrow \mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_2 | E_1) \cdot \mathbb{P}(E_1)$$

## CHAIN RULE

- ▶ The chain rule for two events:

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_2 | E_1) \cdot \mathbb{P}(E_1) = \mathbb{P}[E_1 | E_2] \mathbb{P}[E_2]$$

- ▶ Can generalize to three or more events:

$$\mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(E_3 | E_2 \cap E_1) \cdot \mathbb{P}(E_2 | E_1) \cdot \mathbb{P}(E_1)$$

- ▶ Next time: law of total probability and Bayes Theorem